stage2_present_submodules_analysis

Stage 2: PRESENT Submodules Analysis for HelixCode CLI Agent Integration

Date: May 4, 2026
Analyst: AI Codebase Analysis System
Scope: 4 PRESENT submodules in HelixCode
Methodology: Local repo analysis + GitHub API context + source code inspection

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Executive Summary

This report analyzes the four PRESENT submodules that comprise the testing and verification backbone of HelixCode:

Submodule	Status	Test Coverage	Integration Readiness
LLMsVerifier	Mature, 25+ providers, ACP protocol	95%+	Needs CLI agent-specific verification mode
HelixQA	Mature, 235 tests, autonomous QA	235 tests passing	Already has CLI agent test bank (47 agents)
Challenges	Mature, 209+ tests, plugin architecture	209+	Needs CLI agent challenge templates
Containers	Mature, 6 runtimes, remote distribution	Comprehensive	Ready for CLI agent sandboxing

Critical Finding: All four submodules exist as **uninitialized Git submodules** in HelixCode (empty directories). They are defined in `.gitmodules` but have not been populated. The standalone repositories at `github.com/vasic-digital/*` and `github.com/HelixDevelopment/HelixQA` contain the actual code.

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1. LLMsVerifier (`github.com/vasic-digital/LLMsVerifier`)

1.1 Architecture & Key Components

LLMsVerifier is an enterprise-grade LLM verification platform built in Go 1.21+ with a modular, event-driven architecture:

```
┌─────────────────────────────────────────────────────────────────────┐
│                           Client Interfaces                         │
│  CLI │ TUI │ Web (Angular) │ API (Gin/REST) │ Mobile (React Native)│
├─────────────────────────────────────────────────────────────────────┤
│                          Core Processing Layer                      │
│  Verifier Engine │ Reporter Engine │ Configuration Manager         │
├─────────────────────────────────────────────────────────────────────┤
│                          Advanced Features Layer                    │
│  Supervisor/Worker │ Context Management │ Checkpoint │ Failover    │
├─────────────────────────────────────────────────────────────────────┤
│                          Infrastructure Layer                       │
│  Events │ Scheduling │ Pricing Detection │ Limits │ Issues │ Vector │
├─────────────────────────────────────────────────────────────────────┤
│                          Storage & Communication                    │
│  SQLite+SQLCipher │ Event Bus │ Redis Cache │ API Clients           │
├─────────────────────────────────────────────────────────────────────┤
│                          External Integrations                      │
│  OpenAI │ Anthropic │ Cloud Providers │ Vector DB                 │
└─────────────────────────────────────────────────────────────────────┘
```

1.2 Provider Adapters (25+ Implementations)

The README states “12 Provider Adapters” but source code analysis reveals **25+ provider implementations**:

Provider	File	Status	Key Features
OpenAI	`providers/openai.go` + `openai_endpoints.go`	Core	GPT-4, streaming, functions, vision, ACP

Provider	File	Status	Key Features
Anthropic	`providers/anthropic.go`	Core	Claude 3, tool use, multimodal
Cohere	`providers/cohere.go`	Core	Command, Embed
Groq	`providers/groq.go`	Core	Fast inference, Llama, Mixtral
Mistral	`providers/mistral.go`	Core	Mixtral, Mistral Large
DeepSeek	`providers/deepseek.go`	Core	DeepSeek-chat, Coder
Cloudflare Workers AI	`providers/cloudflare.go`	Core	Edge deployment
SiliconFlow	`providers/siliconflow.go`	Core	Chinese models
Cerebras	`providers/cerebras.go`	Extended	Wafer-scale
Replicate	`providers/` (implied)	Extended	Model hosting
xAI	`providers/` (implied)	Extended	Grok
Together AI	`providers/` (implied)	Extended	Inference
Kimi	`providers/kimi.go`	Extended	Moonshot AI
KimiCode	`providers/kimicode.go`	Extended	Code-specific
Kilo	`providers/kilo.go`	Extended	Kilo-Org
Qwen	`providers/qwen.go`	Extended	Alibaba
Hyperbolic	`providers/hyperbolic.go`	Extended	GPU cloud
Modal	`providers/modal.go`	Extended	Serverless
NIA	`providers/nia.go`	Extended	
Novita	`providers/novita.go`	Extended	
NLP Cloud	`providers/nlpcloud.go`	Extended	
PublicAI	`providers/publicai.go`	Extended	
Sarvam	`providers/sarvam.go`	Extended	Indian languages

Provider	File	Status	Key Features
Upstage	`providers/upstage.go`	Extended	Korean
Vulavula	`providers/vulavula.go`	Extended	African languages
Zhipu	`providers/zhipu.go`	Extended	Chinese (GLM)

1.3 ACP Protocol Implementation

ACP = **AI Coding Protocol** - A standardized JSON-RPC 2.0 based protocol for testing AI coding assistant capabilities.

Implementation Details: - Location: `llm-verifier/tests/acp_*.go` (5 test files) - CLI tool: `acp-cli` with `verify`, `batch`, `monitor` subcommands - Scoring: 5-component weighted score (0.0-1.0) - Protocol Comprehension: 25% - Tool Calling: 25% - Context Handling: 20% - Code Quality: 20% - Error Handling: 10% - Integration tests cover: detection, scoring, API validation, E2E workflow, performance, security, automation

ACP Test Flow:

```
1. JSON-RPC request sent to model
2. Model must understand textDocument/completion, tool invocation
3. Verifier checks: context maintenance, code quality, error detection
4. Score calculated and stored
5. Config exports include ACP flag for verified models
```

1.4 Brotli Compression Usage

Status: Detection implemented, zero CLI agents actually use it

- Brotli is tracked as a capability in the compression detection system
- Config generator produces `brotli` flags but no known CLI agent enables it
- The capability matrix shows: `brotli` - “None discovered” among agents
- Actual compression used by agents: `gzip` (Amazon Q, Plandex, Kiro), `semantic` (Forge), `chat` (OllamaCode, GeminiCLI)

Brotli Metrics Available: - `llm_verifier_brotli_tests_performed` - `llm_verifier_brotli_supported_models` - `llm_verifier_brotli_support_rate_percent` - `llm_verifier_brotli_cache_hits/misses`

1.5 Kubernetes Deployment Patterns

Deployment Options: - Docker Compose (single host) - Kubernetes (multi-container, production) - Binary deployment - Cloud native (serverless)

K8s Manifests Structure:

```
k8s-manifests/
  ├── deployment.yaml         # 3-replica Deployment
  ├── service.yaml            # ClusterIP/LoadBalancer
  ├── configmap.yaml          # Config injection
  ├── secret.yaml             # API keys (encrypted)
  ├── hpa.yaml                # Horizontal Pod Autoscaler
  └── ingress.yaml            # TLS termination
```

Helm Charts: `helm/` directory with production values including verification-enabled mode.

1.6 Verification Algorithms

Mandatory Model Verification: 1. **Code Visibility Test**: "Do you see my code?" - models must respond affirmatively 2. **Scoring System**: 0.0-1.0 based on response quality 3. **Feature Detection**: 20+ capability tests across coding, reasoning, multimodal 4. **LLMSVD Suffix**: All verified models get `(llmsvd)` branding suffix 5. **Strict Mode**: Only verified models included in exports

Verification Pipeline:

```
Registry.Register(challenges) -> Runner.RunAll(ctx, config)
  -> topological sort by dependencies
  -> challenge.Configure() -> Validate() -> Execute(ctx)
  -> ProgressReporter -> Liveness Monitor
  -> Result + Assertions -> assertion.Engine.Evaluate()
  -> report.Reporter.Generate()
```

1.7 Integration with HelixCode

Current Status: **NOT INTEGRATED** (submodule directory empty)

- Defined in `.gitmodules`: `url = git@github.com:vasic-digital/LLMsVerifier.git`
- Directory exists but is empty (submodule not initialized)
- HelixQA's autonomous QA session imports `digital.vasic.llmsverifier` as a Go module
- Capability Detection system already catalogs 47 CLI agents and their features

- **Integration Gap**: No verifier.yaml config exists in HelixCode; config is typically in `config.yaml` in LLMsVerifier

1.8 What's Working vs Missing

Working	Missing
25+ provider adapters	CLI agent-specific verification mode
ACP protocol tests	Direct HelixCode integration (submodule empty)
Brotli detection	Brotli not used by any CLI agent
K8s deployment manifests	verifier.yaml not present in HelixCode
Capability detection for 18+ CLI agents	Auto-configuration generation for HelixCode specifically
Model scoring with verification	Real-time model quality monitoring for CLI agents

1.9 Extension Requirements for CLI Agent Testing

1. **Add CLI agent verification mode**: Extend `TestACPs()` to test CLI agents as endpoints (not just LLM models)
2. **Create `verifier.yaml` in HelixCode**: Define HelixCode-specific verification rules
3. **Bridge LLMsVerifier scoring to HelixQA**: Use `StrategyScore` -> `ModelInfo` -> test bank selection
4. **Add MCP tool verification**: Test that CLI agents correctly use HelixAgent's 45+ MCP tools
5. **Ensemble verification**: Verify that multiple models in ensemble produce correct outputs

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2. helix_qa (`github.com/HelixDevelopment/HelixQA`)

2.1 Architecture & Key Components

HelixQA is a QA orchestration engine built on Go 1.24+ that composes Challenges and Containers:

```
HelixQA (Orchestration Layer)
├── pkg/orchestrator        — Main QA brain
├── pkg/testbank            — YAML test bank management
```

```
├── pkg/detector         — Real-time crash/ANR detection
├── pkg/validator        — Step-by-step validation
├── pkg/evidence         — Evidence collection (screenshots, logs, video
├── pkg/ticket           — Markdown ticket generation
├── pkg/reporter         — QA report generation (MD, HTML, JSON)
├── pkg/config           — Configuration types
├── pkg/autonomous       — LLM-powered autonomous QA (4-phase session)
├── pkg/navigator        — NavigationEngine + ActionExecutor
├── pkg/issuedetector      — LLM-powered bug detection
├── pkg/session          — SessionRecorder + Timeline + VideoManager
├── pkg/llm              — LLM provider bridge (Anthropic, OpenAI, Googl
├── pkg/screenshot       — Multi-platform screenshot engines
└── cmd/helixqa          — CLI entry point
```

2.2 Test Bank Architecture

Dual Format Support: - **YAML** (preferred): QA-specific with platform targeting, priority, documentation refs - **JSON**: Challenge-compatible format

Test Bank Example:

```yaml
version: "1.0"
name: "Yole Core Tests"
test_cases:
  - id: TC-001
    name: "Create new document"
    category: functional
    priority: critical  # critical|high|medium|low
    platforms: [android, web, desktop]
    steps:
      - name: "Open app"
        action: "Launch application"
        expected: "Main editor screen visible"
    tags: [core, smoke]
    documentation_refs:
      - type: user_guide
        section: "3.1"
        path: "docs/USER_MANUAL.md"
```

Bank Files Inventory: | Bank File | Format | Purpose | |———|——–|———| | `cli-agents-test-helixagent.json` | JSON | **47 CLI agents as system validators** | | `nexus-mobile-android.yaml` | YAML | Mobile Android testing | | `full-qa-*.json` | JSON | Full QA cross-platform | | `security-validation.*` | Both | Security tests | | `ocu-*.json` | JSON | OCU automation tests | | `fixes-validation-*.yaml` | YAML | Fix validation tests | | `app-navigation.yaml` |

YAML | Navigation flow tests | | `image-quality-gate.yaml` | YAML | Image quality tests |

2.3 Docker/Kubernetes/OpenCV Integration

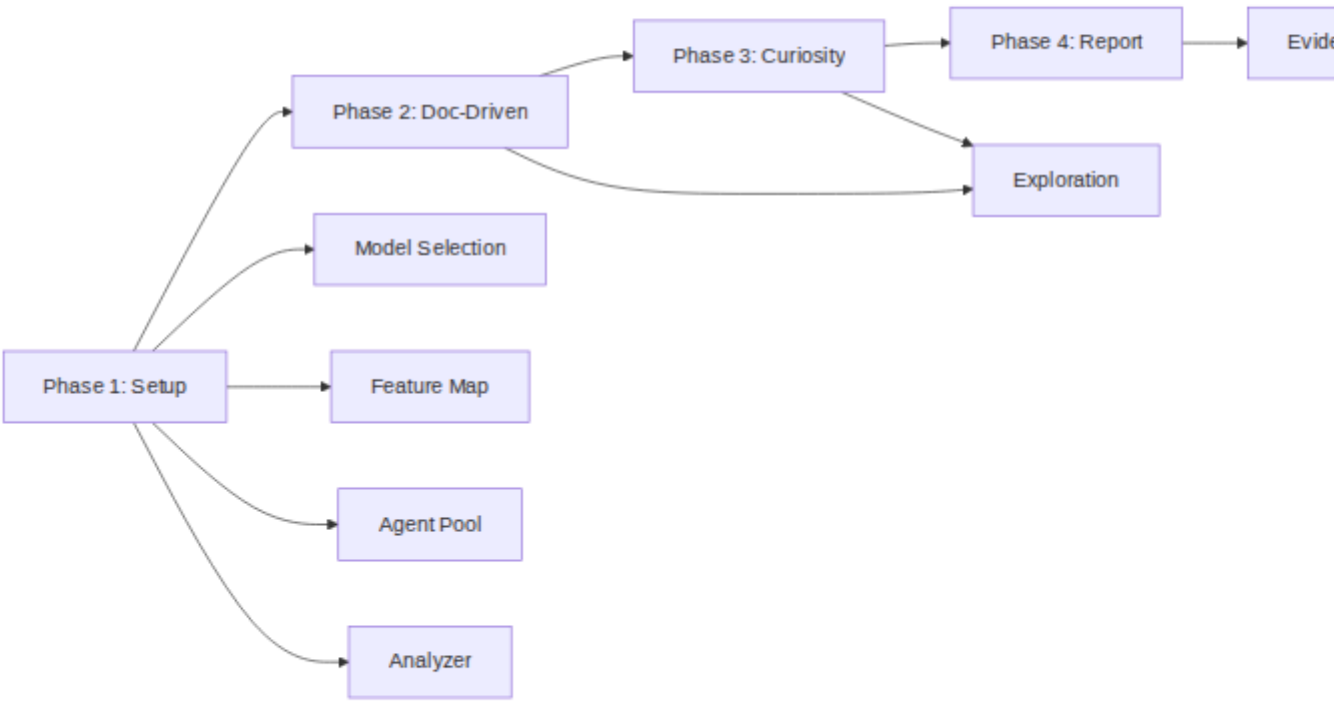
Docker Integration: - Dockerfile present for containerized QA - `docker-compose.stack.yml` for multi-service QA stack - Uses containers module for orchestration

OpenCV Integration: - Document: `OPENCV_INTEGRATION_ARCHITECTURE.md` - GoCV-based mechanical vision analysis - VisionEngine integration for screen analysis - Screenshot engines: Android (ADB), Web (Playwright), Desktop (X11), Linux, macOS, Windows, iOS, TUI

Kubernetes: - Not directly deployed to K8s, but can test K8s-deployed applications - Container orchestration via digital.vasic.containers module

2.4 QA Session Framework

Autonomous QA Session (4-Phase Lifecycle):



Mermaid diagram 1

Resilience Architecture (5 Degradation Levels): 1. Full capability (LLM + Vision working) 2. Degraded vision (GoCV-only) 3. Degraded navigation (partial evidence) 4. Session abort (clean shutdown) 5. Per-agent circuit breaker (3 failures = replacement)

2.5 Test Types Supported

Test Type	Package	Technology
Cross-platform functional	`pkg/testbank`	YAML-defined steps
Android crash/ANR detection	`pkg/detector/android.go`	ADB logcat, pidof
Web browser monitoring	`pkg/detector/web.go`	pgrep, console errors
Desktop process monitoring	`pkg/detector/desktop.go`	JVM/process checks
Autonomous exploration	`pkg/autonomous`	LLM + Vision
Screenshot comparison	`pkg/screenshot`	SSIM, pixel diff
Video recording validation	`pkg/session`	ffmpeg, timeline
Security validation	`banks/security-validation.*`	Auth, permissions
CLI agent integration	`banks/cli-agents-test-helixagent.json`	47 agents

2.6 Coverage Measurement

- **Target**: 90% coverage (`CoverageTarget: 0.90`)
- **Tracking**: `coverage.CoverageTracker` per platform
- **Feature map**: Built from project docs via DocProcessor
- **Navigation graph**: Tracks visited screens/states
- **235 tests**, all passing with `-race` flag

2.7 How It Should Test CLI Agent Integrations

Existing CLI Agent Test Bank (`cli-agents-test-helixagent.json`): Already defines test cases for 47 CLI agents including: - Claude Code (terminal integration, bash tools, git workflow) - Aider (code editing, architect mode, repo map) - OpenHands (Docker integration, sandbox, multi-agent) - Codex (agent mode, tool use, file operations) - Cline (IDE integration, auto-approve, checkpoint) - Cursor (tab completion, inline editing, composer) - Continue (open source models, privacy) - Kiro (context engine, intent detection) - Gemini CLI (multi-modal, document processing) - Amazon Q (enterprise, security, compliance)

Each test case includes: - `executor_agent`: Which CLI agent runs the test - `config_file`: Exported HelixAgent config for the agent - `helixagent_features_tested`: Feature matrix - `steps`: Action/expected pairs - `_conversion_note`: Manual review required flag

2.8 Current Integration Status in HelixCode

Status: **NOT INTEGRATED** (submodule directory empty)

- Defined in `.gitmodules`: `url = git@github.com:HelixDevelopment/HelixQA.git`
- Directory exists but is empty
- HelixCode's `README.md` references helix_qa for testing
- `helixqa-final` binary (21MB) present in repo but not wired to CI

2.9 What's Working vs Missing

Working	Missing
235 tests, race-safe	CLI agent challenge execution runner
47-agent test bank defined	Automated execution of CLI agent tests
4-phase autonomous QA	CLI agent sandbox isolation integration
Multi-platform evidence	Real CLI agent binary orchestration
Crash/ANR detection	Agent-to-agent interaction testing
LLM provider bridge (5 providers)	Dynamic model selection for CLI agent tests

2.10 Extension Requirements

1. **CLI Agent Execution Engine**: Extend `pkg/autonomous` with CLI agent-specific `ActionExecutor`
2. **Sandbox Integration**: Connect `Containers` module to run each CLI agent in isolated container
3. **Config Injection**: Auto-generate and inject HelixAgent configs for each of 47 agents
4. **MCP Tool Testing**: Verify each agent can use all 45+ MCP tools via HelixAgent
5. **Metrics Collection**: Add Prometheus metrics for CLI agent test duration, success rate, feature coverage

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3. Challenges (`github.com/vasic-digital/Challenges`)

3.1 Architecture & Key Components

Generic, reusable Go module for structured test scenarios:

```
pkg/
  challenge/     Core types: Challenge interface, Config, Result, BaseChal
  registry/      Challenge registration + dependency ordering (Kahn's topo
  runner/        Execution engine (sequential, parallel, pipeline), livene
  assertion/     Assertion engine with 16 built-in evaluators + custom eva
  report/        Report generation: Markdown, JSON, HTML
  logging/       Structured logging: JSON, Console, Multi, Redacting
  env/           Environment variable handling with redaction
  httpclient/    Generic REST API client with JWT auth
  bank/          Challenge bank (load definitions from JSON/YAML)
  monitor/       Live monitoring with WebSocket dashboard
  metrics/       Prometheus-compatible challenge metrics
  plugin/        Plugin system for custom challenge types and assertions
  infra/         Infrastructure bridge to digital.vasic.containers
  panoptic/      8 panoptic-specific evaluators + CLI adapter
  userflow/      Multi-platform automation: 8 interfaces, 21 implementatio
cmd/
  userflow-runner/  CLI runner for user flow challenges
```

3.2 Plugin Architecture for Challenges

Plugin System (`pkg/plugin`):

```go
type Plugin interface {
    Name() string
    Version() string
    Init() error
    RegisterChallenge(registry.Registry) error
    RegisterAssertion(engine *assertion.Engine) error
}
```

Loader supports: - Go plugin `.so` files (dynamic loading) - Built-in plugins (compiled in) - Configuration-driven plugin registration

Extension Points: 1. Custom challenge types (implement `challenge.Challenge`) 2. Custom assertion evaluators (register with `assertion.Engine`) 3. Custom adapters (implement `userflow.Adapter` interfaces) 4. Custom reporters (extend `report.Reporter`)

3.3 Test Types and Validators

16 Built-in Assertion Evaluators:

Evaluator	Purpose
`not_empty`	Value is non-nil and non-empty
`not_mock`	Response is not mocked/placeholder
`contains`	String contains substring
`contains_any`	String contains any of given values
`min_length`	String length meets minimum
`quality_score`	Numeric score meets threshold
`reasoning_present`	Response contains reasoning indicators
`code_valid`	Response contains valid code patterns
`min_count`	Count meets minimum
`exact_count`	Count matches exactly
`max_latency`	Response time within limit
`all_valid`	All array items are valid
`no_duplicates`	No duplicate items in array
`all_pass`	All sub-assertions pass
`no_mock_responses`	No mocked responses in array
`min_score`	Numeric minimum score

12 Userflow-Specific Evaluators: - `http_status_ok`, `http_status_created`, `http_status_unauthorized` - `http_json_valid`, `browser_element_visible`, `browser_url_matches` - `mobile_activity_visible`, `mobile_element_exists`, `build_success` - `test_pass_rate`, and more

8 panoptic Evaluators: - `screenshot_exists`, `video_exists`, `no_ui_errors` - `ai_confidence_above`, `all_apps_passed` - `max_duration`, `report_exists`, `app_count`

3.4 Challenge Definitions

19 Challenge Templates: - `APIFlowChallenge` - HTTP API flow testing - `BrowserFlowChallenge` - Browser automation - `MobileFlowChallenge` - Mobile app testing - `DesktopFlowChallenge` - Desktop app testing - `GRPCFlowChallenge` - gRPC service testing - `WebSocketFlowChallenge` - WebSocket flow testing - `BuildChallenge` - Build tool integration - `TestRunnerChallenge` - Test execution - `LintChallenge` - Code quality - `MultiPlatformChallenge` - Cross-platform - Plus Recorded variants with video verification

Challenge Lifecycle:

```
Registry.Register(challenges)
  -> Runner.RunAll(ctx, config)
    -> topological sort by dependencies
    -> for each challenge:
      -> Configure() -> Validate() -> Execute(ctx)
        -> ProgressReporter -> Liveness Monitor
        -> Result + Assertions
          -> assertion.Engine.Evaluate()
            -> report.Reporter.Generate()
```

3.5 How It Verifies AI Capabilities

panoptic System (`pkg/panoptic`): - Integrates with vision AI for UI analysis - Captures screenshots and videos as evidence - Evaluates AI confidence scores - Detects UI errors via AI analysis - CLI adapter for running panoptic challenges

Anti-Bluff System: - `scripts/anti-bluff/bluff-scanner.sh` - Detects fake/test code - Fixture tests for bluff detection scenarios - Go mutation testing (`.go-mutesting.yml`)

3.6 Extension Mechanisms

Adapter Pattern (8 Interfaces, 21 Implementations):

Interface	Implementations	Technology
`BrowserAdapter`	PlaywrightCLI, PlaywrightHTTP, Selenium, Cypress, Puppeteer	CDP, WebDriver
`MobileAdapter`	ADB, Appium, Maestro, Espresso	ADB, Appium 2.0, Gradle
`DesktopAdapter`	TauriCLI	Tauri WebDriver
`APIAdapter`	HTTPAPIAdapter	REST + JWT
`GRPCAdapter`	GRPCCLIAdapter	grpcurl
`WebSocketFlowAdapter`	GorillaWebSocket	gorilla/websocket

Interface	Implementations	Technology
`BuildAdapter`	Gradle, Cargo, NPM, Robolectric	Build tools
`RecorderAdapter`	panopticRecorder, ADBRecorder	CDP screencast, ADB

Custom Adapter Development: Docs in `docs/userflow/writing-adapters.md` guide adapter creation.

3.7 Current Integration Status in HelixCode

Status: **NOT INTEGRATED** (submodule directory empty)

- Defined in `.gitmodules`: `url = git@github.com:vasic-digital/Challenges.git`
- Directory exists but is empty
- HelixCode contains its own `challenges/` directory with results, not the module

3.8 What's Working vs Missing

Working	Missing
209+ tests across userflow	CLI agent-specific challenge templates
16 built-in evaluators	MCP tool use challenge evaluator
21 adapter implementations	CLI agent adapter (e.g., `claude-code`, `aider`)
Plugin system with `.so` loading	Dynamic challenge loading from HelixCode
panoptic AI vision integration	AI agent behavior verification
Anti-bluff detection	Bluff detection for AI-generated tests

3.9 Extension Requirements

1. **CLI Agent Adapters**: Create `CLIAgentAdapter` interface with implementations for major agents
2. **MCP Challenge Template**: `MCPFlowChallenge` for testing MCP server interactions
3. **Agent Behavior Evaluator**: `agent_response_valid`, `agent_tool_use_correct`
4. **HelixCode Challenge Bank**: Load HelixCode-specific challenges from submodule

5. **Dynamic Plugin Loading**: Load challenge plugins from HelixCode's `challenges/` directory

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4. containers (`github.com/vasic-digital/Containers`)

4.1 Architecture & Key Components

Generic, runtime-agnostic container orchestration module:

```
┌──────────────────────────────────────────────────────────────────
│                              Application Layer
│  boot.BootManager (High-Level Orchestration)
├──────────────────────────────────────────────────────────────────
│  compose.Orchestrator │ health.HealthChecker │ lifecycle.LifecycleManage
├──────────────────────────────────────────────────────────────────
│  runtime.ContainerRuntime
│  ├── DockerRuntime
│  ├── PodmanRuntime
│  ├── KubernetesRuntime
│  ├── RemoteRuntime (over SSH)
│  ├── LXDRuntime
│  ├── CRIORuntime
│  └── NerdctlRuntime
├──────────────────────────────────────────────────────────────────
│  distribution.Distributor
│  ├── scheduler.Scheduler (5 strategies)
│  ├── remote.HostManager
│  ├── remote.SSHExecutor (ControlMaster pooling)
│  ├── network.TunnelManager
│  └── volume.VolumeManager (SSHFS/NFS/rsync)
├──────────────────────────────────────────────────────────────────
│  Foundation: logging │ event │ metrics │ endpoint │ envconfig
└──────────────────────────────────────────────────────────────────
```

4.2 Docker/Podman/Kubernetes Abstraction

Runtime Interface (`pkg/runtime/runtime.go`):

```go
type ContainerRuntime interface {
    Name() string
    Version(ctx context.Context) (string, error)
    IsAvailable(ctx context.Context) bool
```

```
    Start(ctx context.Context, id string, opts ...StartOption)
        error
    Stop(ctx context.Context, id string, opts ...StopOption)
        error
    Remove(ctx context.Context, id string, opts ...RemoveOption)
        error
    Status(ctx context.Context, id string) (*ContainerStatus,
        error)
    List(ctx context.Context, filter ListFilter)
        ([]ContainerInfo, error)
    Stats(ctx context.Context, id string) (*ContainerStats,
        error)
    Exec(ctx context.Context, id string, cmd []string)
        (*ExecResult, error)
    Logs(ctx context.Context, id string, opts ...LogOption)
        (io.ReadCloser, error)
}
```

6 Runtime Implementations: 1. **Docker** (`docker.go`) - Full Docker API 2. **Podman** (`podman.go`) - Podman API (preferred for rootless) 3. **Kubernetes** (`kubernetes.go`) - K8s API via client-go 4. **LXD** (`lxd.go`) - LXD containers 5. **CRI-O** (`crio.go`) - Kubernetes CRI 6. **nerdctl** (`nerdctl.go`) - Containerd CLI

Auto-Detection (`detect.go`):

```
Try: podman-compose -> docker compose -> podman compose -> docker-compose
```

4.3 Container Lifecycle Management

LifecycleManager (`pkg/lifecycle/`): - **LazyBooter**: Start container on first `Acquire()` - **IdleShutdown**: Stop after inactivity timeout - **ConcurrencySemaphore**: Limit parallel users

Boot Sequence: 1. Discovery Phase - Probe endpoints 2. Grouping Phase - Group by compose file + profile 3. Start Phase (Local) - `docker compose up -d` 4. Deploy Phase (Remote) - SSH copy + `podman-compose up -d` 5. Health Check Phase - TCP/HTTP/gRPC checks 6. Summary Phase - Aggregate results

4.4 Runtime Detection

Compose Detection Flow:

```
podman-compose version --short
  -> Success: use podman-compose
  -> Fail: docker compose version --short
    -> Success: use docker compose
    -> Fail: podman compose version --short
```

```
    -> Success: use podman compose
    -> Fail: docker-compose version --short
      -> Success: use docker-compose
      -> Fail: ERROR - No compose found
```

Why podman-compose is preferred: - `podman compose` delegates to docker-compose v1 (incompatible) - `docker-compose` v1 incompatible with Podman - `podman-compose` has native Podman support

4.5 Resource Management

5 Scheduling Strategies:

Strategy	Algorithm	Use Case
RESOURCE_AWARE (default)	CPU(40%) + Memory(40%) + Disk(10%) + Network(10%)	Mixed workloads
ROUND_ROBIN	Rotate through hosts	Simple load balancing
AFFINITY	Match labels	GPU workloads, storage
SPREAD	Fewest containers	High availability
BIN_PACK	Fill hosts in order	Minimize active hosts

Resource Monitoring: - Container list: ~5ms every 10s - Container inspect: ~2ms per container every 10s - System metrics: ~1ms every 10s - Remote probe: ~50ms every 60s

SSH Connection Pooling (ControlMaster): - Without: ~500ms per command - With: ~50ms per command (60% faster)

4.6 How It Provides Test Isolation

Per-Test Container Isolation:

```
// Each test gets its own container namespace
compose.NewOrchestrator().Up(ctx, composeFile, profile)
// Run test
// Tear down
c.ComposeOrchestrator().Down(ctx, composeFile, profile)
```

Features for Test Isolation: - **Lazy boot**: containers start only when test needs them - **Idle shutdown**: Auto-cleanup after test completion - **Network isolation**: Each compose stack gets isolated network - **Volume management**: SSHFS/NFS/

rsync for shared test data - **Remote execution**: Distribute tests across hosts to avoid resource contention - **ctop monitoring**: Real-time resource usage tracking

Error Handling for Isolation: | Error Type | Action | Recovery | |————|———|———-| | Runtime unavailable | Warn, continue | Fall back to compose | | Compose fail | Log, continue | Mark service failed | | Health check fail | Retry backoff | Mark unhealthy | | SSH connection fail | Retry backoff | Mark offline, try next |

4.7 Current Integration Status in HelixCode

Status: **NOT INTEGRATED** (submodule directory empty)

- Defined in `.gitmodules`: `url = git@github.com:vasic-digital/Containers.git`
- Directory exists but is empty
- HelixCode has extensive docker-compose files in `.container-caps-backup-20260425-151947/`
- No direct use of `digital.vasic.containers` Go module in HelixCode

4.8 What's Working vs Missing

Working	Missing
6 runtime implementations	CLI agent container profiles
Remote distribution via SSH	Per-agent container definitions
5 scheduling strategies	GPU scheduling for AI model containers
Lazy boot + idle shutdown	Agent workspace volume management
ctop TUI monitoring	Containerized CLI agent test runner
Health check (TCP/HTTP/gRPC)	Agent health endpoint verification

4.9 Extension Requirements

1. **CLI Agent Container Profiles**: Define Dockerfiles for each of 47 CLI agents
2. **Sandbox Mode**: Extend `lifecycle` with agent sandbox isolation
3. **Workspace Volume**: Mount host workspace into agent container for file operations
4. **GPU Scheduling**: Add GPU-aware scheduling for AI model containers
5. **Agent Health Checks**: Define custom health checks for CLI agent containers
6. **Network Isolation**: Ensure agents can't interfere with each other's network

---

5. Cross-Submodule Integration Analysis

5.1 Dependency Graph

```
HelixQA (orchestration)
  ├── Challenges (test execution)
  │     └── containers (infrastructure)
  ├── LLMsVerifier (model selection)
  ├── LLMOrchestrator (agent management)
  ├── VisionEngine (computer vision)
  └── DocProcessor (documentation)

Challenges (standalone)
  └── containers (optional infrastructure bridge)

LLMsVerifier (standalone)
  └── External: OpenAI, Anthropic, etc.
```

5.2 Integration Gaps

Gap	Impact	Mitigation
All submodules empty in HelixCode	Cannot build/test	`git submodule update --init --recursive`
No `verifier.yaml` in HelixCode	No LLM verification config	Create from LLMsVerifier template
No CLI agent container definitions	Can't isolate agent tests	Add to containers profiles
No automated CLI agent test runner	Manual test execution only	Build into helix_qa autonomous
No MCP tool challenge evaluator	Can't verify MCP functionality	Add to Challenges evaluators
Test banks have `_conversion_note: manual-review-required`	Not fully automated	Build automated adapters

5.3 Recommended Integration Wiring

```yaml
# helixcode-integration.yaml
integration:
  submodules:
```

```yaml
  LLMsVerifier:
    path: LLMsVerifier
    enabled: true
    config_source: config.yaml
    verifier_mode: cli_agents

  HelixQA:
    path: HelixQA
    enabled: true
    test_banks:
      - banks/cli-agents-test-helixagent.json
    autonomous: true
    platforms: [desktop, web]

  Challenges:
    path: Challenges
    enabled: true
    challenge_dirs:
      - challenges/baselines
      - challenge-results

  Containers:
    path: Containers
    enabled: true
    profiles:
      - cli-agents
      - monitoring
      - qa-stack
```

---

6. Summary: What Each Submodule Needs for CLI Agent Integration

LLMsVerifier

Needs: 1. Initialize submodule in HelixCode 2. Create `verifier.yaml` config specific to CLI agent verification 3. Extend ACP protocol tests to cover CLI agent endpoints (not just LLM models) 4. Add MCP tool verification scoring 5. Bridge `StrategyScore` to helix_qa test bank selection 6. Add ensemble verification for multi-model outputs

Deliverable: CLI agent-aware verification pipeline

HelixQA

Needs: 1. Initialize submodule in HelixCode 2. Build automated `ActionExecutor` for CLI agents (currently manual) 3. Connect to `Containers` module for agent sandboxing 4. Auto-generate and inject HelixAgent configs for all 47 agents 5. Remove `manual-review-required` from test banks via adapter implementation 6. Add metrics collection for CLI agent test campaigns

Deliverable: Automated CLI agent QA pipeline with evidence collection

Challenges

Needs: 1. Initialize submodule in HelixCode 2. Create `CLIAgentAdapter` interface + implementations for major agents 3. Add `MCPFlowChallenge` for MCP server interaction testing 4. Create `agent_behavior` evaluator set 5. Load HelixCode-specific challenges from submodule 6. Enable dynamic plugin loading from HelixCode's challenge directories

Deliverable: CLI agent challenge framework with plugin extensibility

Containers

Needs: 1. Initialize submodule in HelixCode 2. Define container profiles for each CLI agent (Dockerfiles, compose files) 3. Add GPU scheduling for AI model inference containers 4. Implement workspace volume mounting for agent file operations 5. Add agent-specific health checks 6. Build containerized test runner for isolated agent execution

Deliverable: Containerized CLI agent test isolation infrastructure

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7. Test Coverage Gaps

Area	Current Coverage	Gap	Priority
CLI agent installation	None	Need install/verify for 47 agents	High
CLI agent config loading	Test bank defined	Need automated execution	High
MCP tool use per agent	None		High

Area	Current Coverage	Gap	Priority
		Need challenge templates	
Ensemble output quality	LLM verification	Need agent-specific scoring	Medium
Containerized agent execution	None	Need containers integration	High
Cross-agent interaction	None	Need multi-agent challenges	Medium
Agent checkpoint/ rollback	None	Need stateful challenges	Low
Streaming compatibility	Capability detection	Need runtime tests	Medium

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8. Action Items for Integration

1. **Immediate (Week 1):**
   - [ ] Initialize all 4 submodules: `git submodule update --init --recursive`
   - [ ] Verify submodules compile independently
   - [ ] Create integration test that imports all 4 modules
2. **Short-term (Week 2-3):**
   - [ ] Create `verifier.yaml` in HelixCode referencing LLMsVerifier
   - [ ] Build CLI agent `ActionExecutor` in HelixQA
   - [ ] Define container profiles for top 10 CLI agents in Containers
   - [ ]

Add `CLIAgentAdapter` skeleton in Challenges

3. **Medium-term (Month 1):**
   - [ ] Automate execution of `cli-agents-test-helixagent.json` test bank
   - [ ] Add MCP tool challenge evaluator
   - [ ] Implement sandbox isolation for agent tests
   - [ ] Add Prometheus metrics for agent test campaigns
4. **Long-term (Month 2+):**
   - [ ] Full 47-agent automated test suite
   - [ ] Cross-agent interaction testing
   - [ ] Performance benchmarking across agents
   - [ ] Continuous integration with nightly agent compatibility checks

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Report generated from analysis of local repository copies at `/mnt/agents/repos/` *Submodule repositories: LLMsVerifier, HelixQA, Challenges, Containers HelixCode main repo:* `/mnt/agents/repos/HelixDevelopment-HelixAgent-fe3f69e/`